

Experiment - Study of LAN Topology, Network devices and Address configuration.

Objective -

- 1.) To study the LAN topology used in college campus.
- 2.) To identify networking device & understand their function.
- 3.) To analyze wired & wireless communications method used in the network.
- 4.) To measure speed of network for wired & wireless connections.
- 5.) To obtain Mac, IP Address & Subnet mask.

• Problem statement - 1

- Draw & identify LAN topology of college campus & clearly label all components such as routes, switch etc.

Soln. The college campus network observed this study consists of:

- * Layer - 3 core switches in the server room.
- * Managed by POE + switches
- * wireless Access point (APs).
- * Ethernet port classroom & laboratories

Problem statement 1

→ Network Topology.

Hybrid Hierarchical topology which combines

i.) Tree Topology.

ii.) Extended star topology.

iii.) Ring Topology (server room)

→ The network hierarchical layer.

① Core layer

② Distribution layer

③ Access layer

→ Core layer

* Core layer contains Server Room and server room have firewall Routers, 2-3rd switches, Patch, Rack, server inverter, etc.

→ Distributed layer

* Distribution layer contain 4 switches.

* the 4 switches are

① Main building → R7 (Room no. 213, 2nd floor)

② Main Building → R1 (Room no. 301, 3rd floor)

③ Main building → R6 (Room no. 113, 1st floor)

iv) Main Build $\rightarrow$ R2 (Room no. - 201, 2nd floor)

* this 4-switches contains -

① 18 port Giga bit Easy smart switch with 16 port ∇ [TL-SG1218MPE]

② (SG3428xMP) $\rightarrow$ 24 port Giga bit & 4-port 10Gb SFP, L2 + managed switch with 24 port POE+.

$\rightarrow$ Now let's discuss these 2 above components -

① TP-Link TL-SG1218MPE

* Likely used for: CCTV camera with access point etc.

* 18 port Gigabit "Fast Smart" switch

* 16 port support POE

② TP-Link Omada SG3428xMP

* Higher-end managed layer 2+ switch.

* 24 POE port, Uplinks, 10Gb SFP.

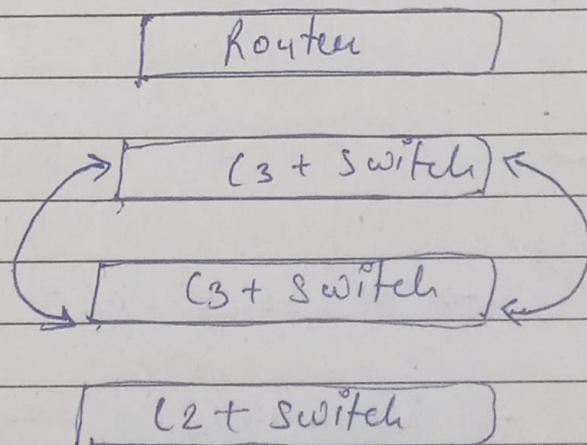
* Usually acts as a main distribution for each other floor

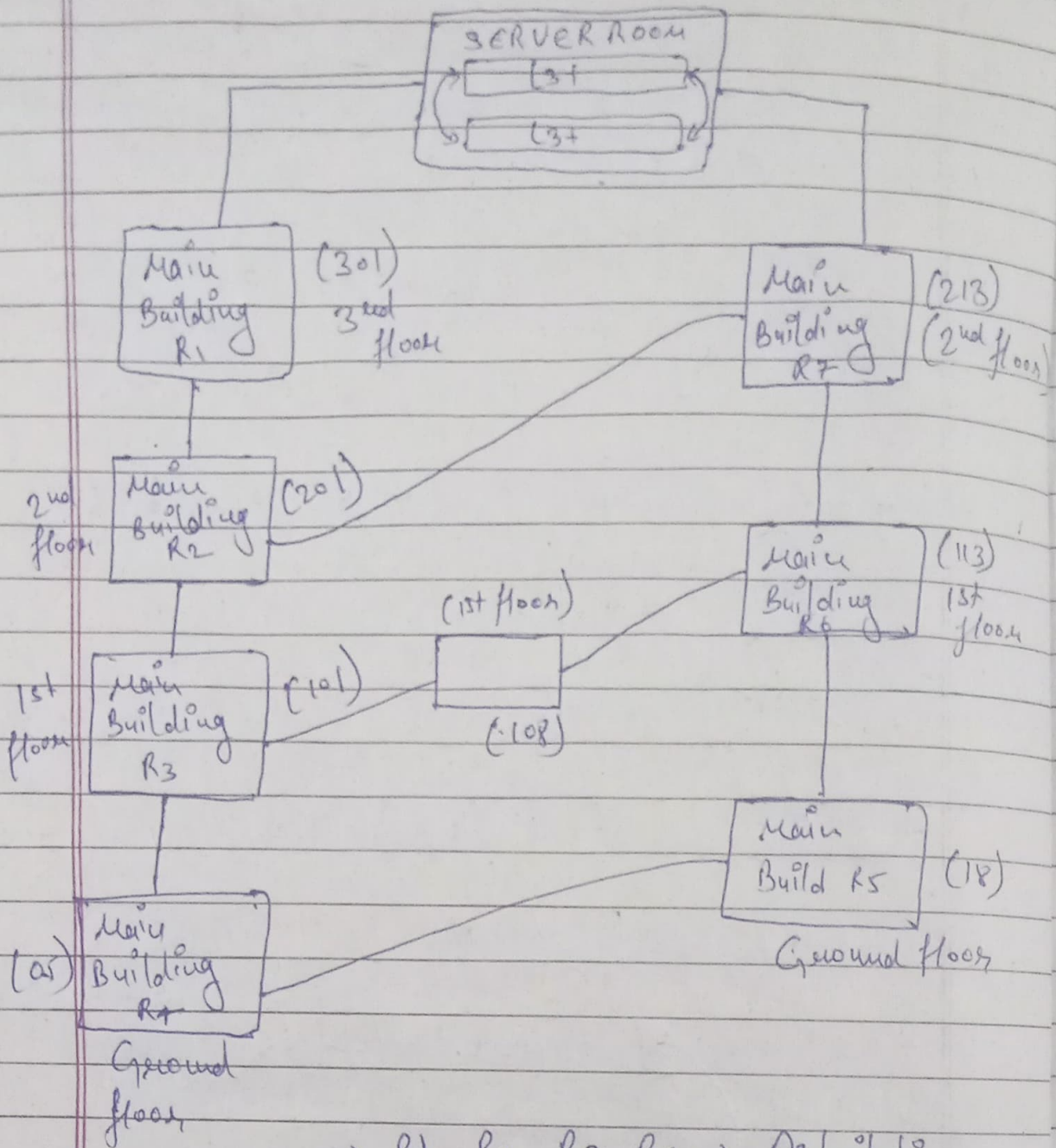
Access layer :-

• Access layer R1 (Room no. - 301, 3rd floor)

- floor) (main Build R1)
 - o Main Build R4 (Room no - 05, Ground floor).
 - o Main Build R5 (Room no. - 18, ground floor)
- * Every switches act as access layer switch helps to provide network connectivity to all classroom, lab, faculty room etc.
- * Access layer directly connect :
 - (1) Ethernet port
 - (100) wireless AP
 - (600) lab system.
 - (10) outdoor wireless AP (ex-08)

→ Server Room Architecture / topology.



Topology.

→ R1, R7, R2, R6 → Distribution layer

→ Server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer.

Microsoft Windows [Version 10.0.26200.8457]

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C:\Users\shwet>netsh wlan show interface

There is 1 interface on the system:

```
Name                : Wi-Fi
Description          : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter
GUID                 : 62719710-9492-41d5-b0d7-0c93b871a0ce
Physical address     : 28:2e:89:cd:ba:9e
Interface type       : Primary
State                : connected
SSID                 : GECK-Capmus-WiFi
AP BSSID             : b0:19:21:74:04:48
Band                 : 2.4 GHz
Channel              : 11
Connected Akm-cipher : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode      : Auto Connect
Receive rate (Mbps)  : 72.2
Transmit rate (Mbps) : 72.2
Signal               : 100%
Rssi                 : -53
Profile              : GECK-Capmus-WiFi
```

Microsoft Windows [Version 10.0.26200.8457]
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```
C:\Users\shwet>ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix  : 
Link-local IPv6 Address . . . . : fe80::a6c5:fe70:dc42:c07d%7
IPv4 Address. . . . .           : 192.168.56.1
Subnet Mask . . . . .           : 255.255.255.0
Default Gateway . . . . .       :
```

Wireless LAN adapter Wi-Fi 3:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

Wireless LAN adapter Wi-Fi:

```

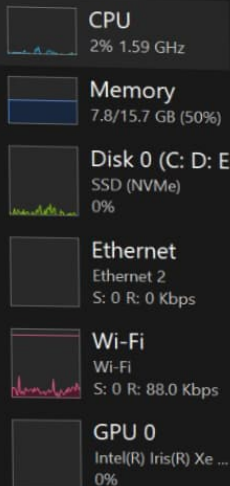
Connection-specific DNS Suffix  : 
IPv6 Address. . . . . : 2409:408a:1507:48c:2c9f:ad31:4c4d:5805
Temporary IPv6 Address. . . . . : 2409:4088:8185:e441:52b:3e21:978b:1243
IPv6 Address. . . . . : 2409:4088:8185:e441:5373:5a14:74b1:9b70
Temporary IPv6 Address. . . . . : 2409:408a:1507:48c:52b:3e21:978b:1243
Link-local IPv6 Address . . . . . : fe80::eafb:35d6:5c2:2848%13
IPv4 Address. . . . . : 10.53.161.4
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::202d:27ff:fedf:5351%13
                          10.53.161.69

```

C:\Users\shwet>

Performance

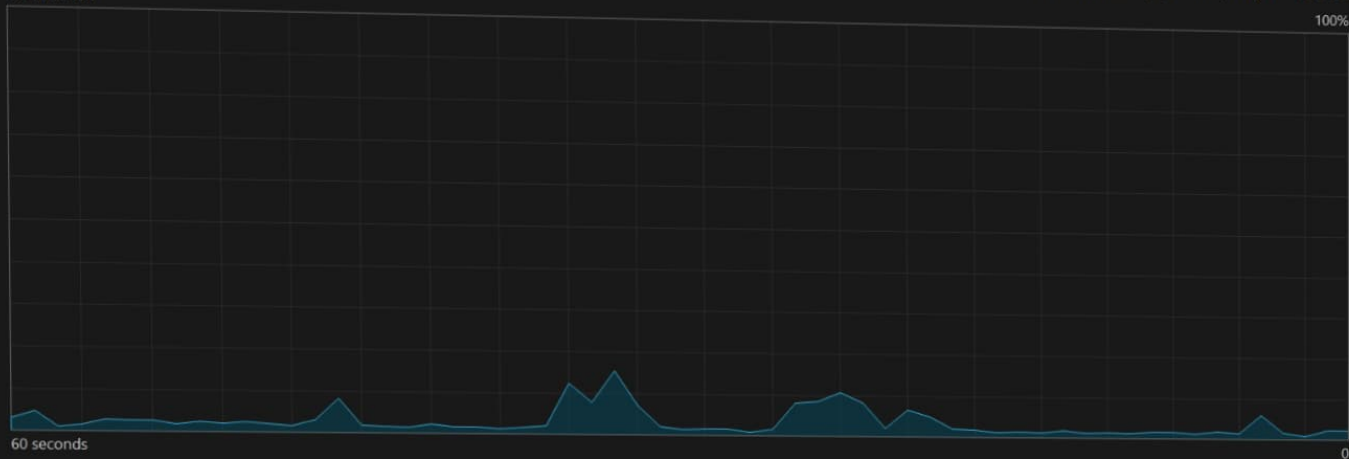
Run new task



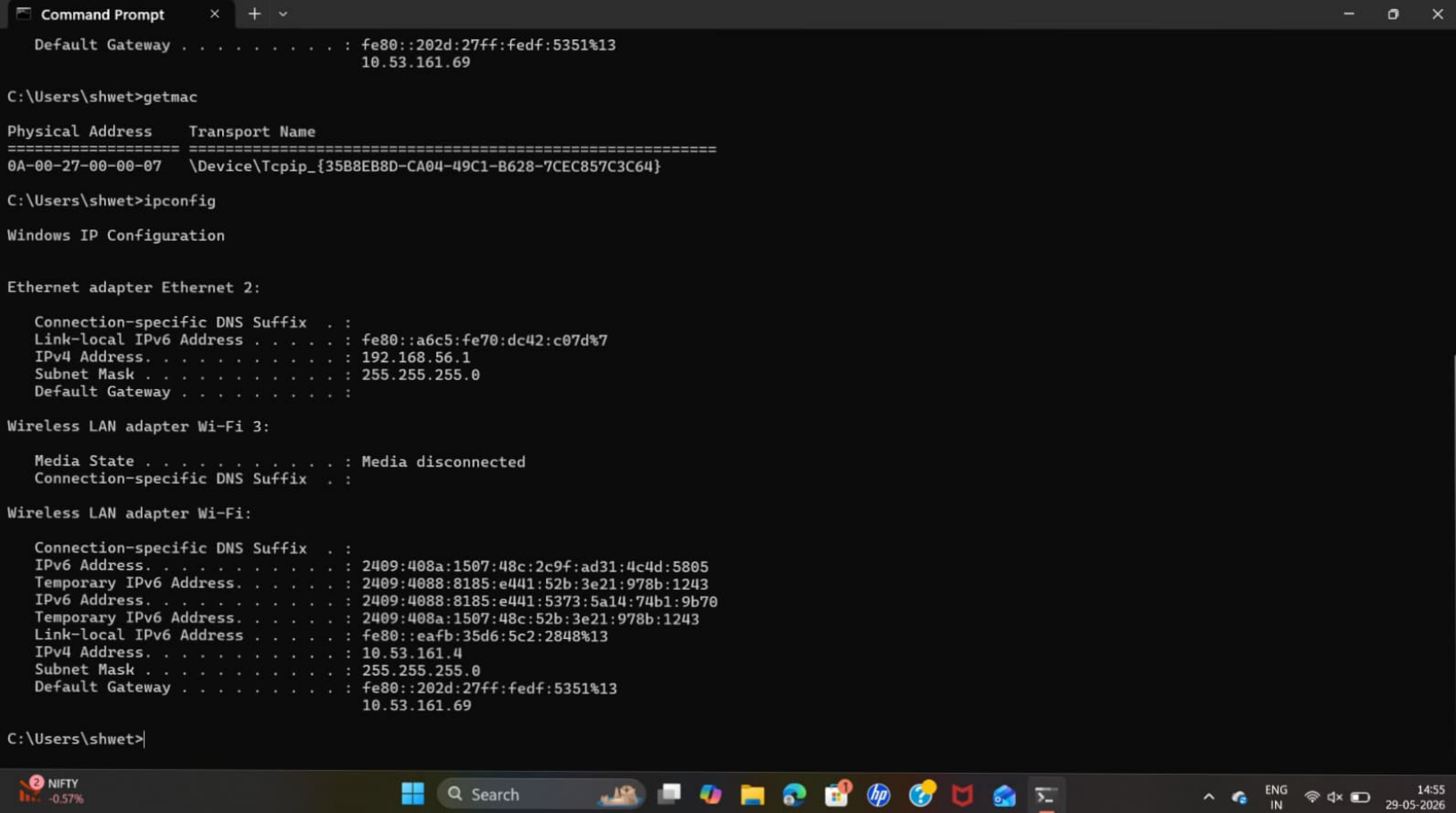
CPU

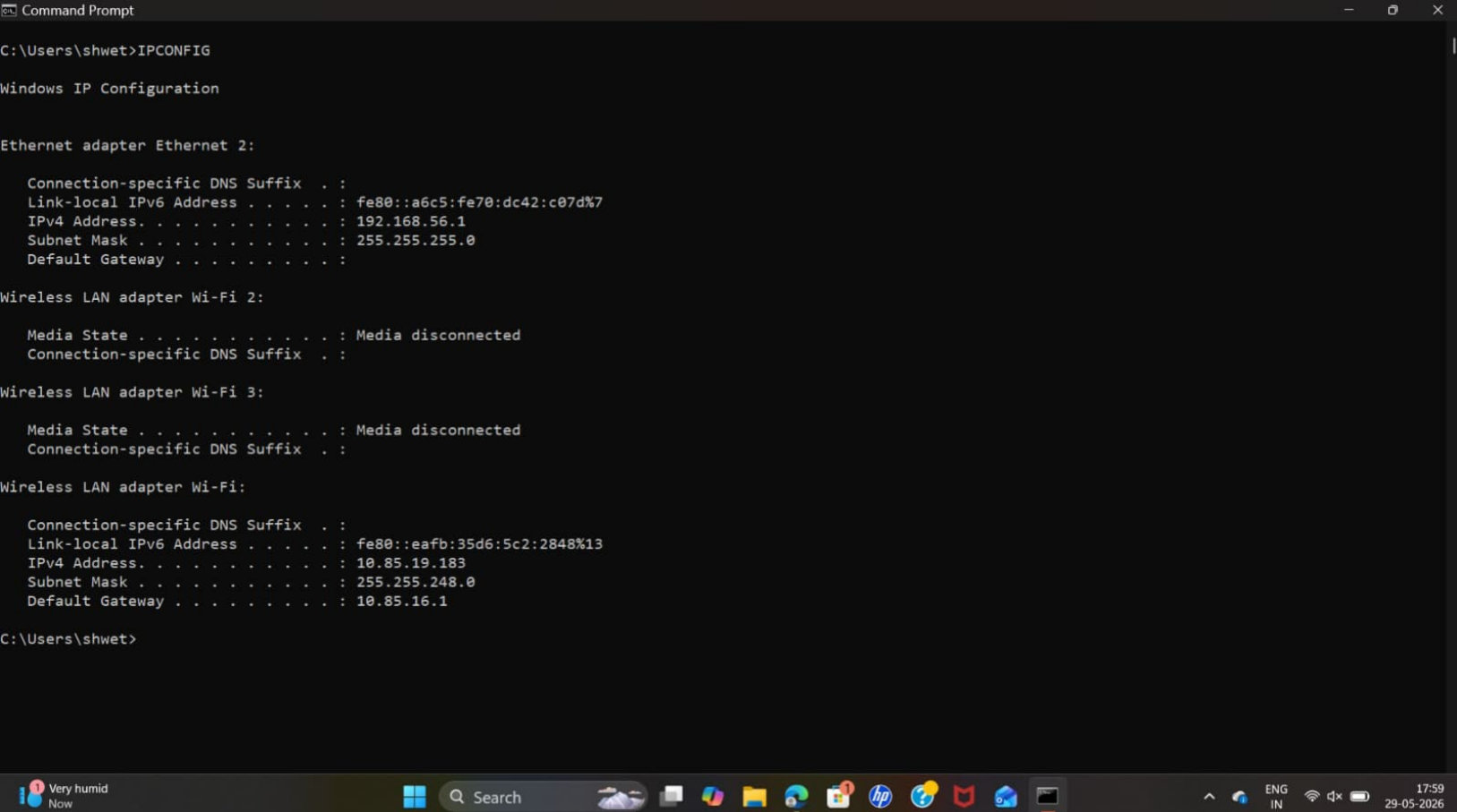
% Utilization

13th Gen Intel(R) Core(TM) i5-1334U



Utilization	Speed	Base speed:	1.30 GHz
2%	1.59 GHz	Sockets:	1
Processes	Threads	Cores:	10
252	3690	Logical processors:	12
Handles		Virtualization:	Enabled
124256		L1 cache:	928 KB
Up time		L2 cache:	6.5 MB
4:07:55:02		L3 cache:	12.0 MB





Difference between wired and wireless speed.

Soln.

<u>wired network.</u>	<u>wireless Network.</u>
→ faster communication	→ slower communication.
→ stable connection.	→ Affected by distance / interference.
→ Low latency.	→ Higher latency.
→ Dedicated physical medium.	→ Shared radio medium.
→ Ethernet provides direct physical communication using network cables resulting in higher speed & stable performance.	
→ Wireless connections use radio waves which are affected by interference, distance obstacles & multiple connected users.	

o Role of subnet mask in network?

Soln.

A Subnet mask identifies the network.

Portion & host portion of an IP address.
It helps devices determine whether another device belongs to the same network or different network.

* It is important for -

- (i) Network segment
- (ii) Efficient routing
- (iii) Communication b/w devices.

Observation,

During the study of the college network infrastructure.

- (1) A centralized server room was used as network backbone.
- (2) Two L3 switches were connected in redundant cyclic structure.
- (3) Managed TP-Link PoE switches were used in network cabinets.

→ Advantages of Existing network

→ S/S

- Scalable architecture
- Centralized management.
- Supports PoE device.
- Easy expansion of floors & classrooms.
- Support both wireless & wired.
- Redundant link for failover.

Conclusion -

The observed campus LAN follow a Hybrid Hierarchical topology consist of TREE Extended star Ring. In this experiment, we studied LAN topology, identified different network devices and learned how to find IP address, MAC address and subnet mask we also compared wired and wireless connection and understood their performance differences.